Tech Stack Breakdown

- **Mobile App Framework:** Flutter (Dart)
- **Backend Database & Authentication:** Supabase (PostgreSQL)
- **AI & Natural Language Processing:** Google Gemini API
- **Admin Web Panel:** Flutter Web

Why These Tools Were Chosen

- **Flutter:** Satisfies the mandatory mobile cross-platform requirement while compiling directly to high-performance native machine code. It compiles UI components consistently across both Android and iOS from a single Dart codebase, saving weeks of development.
- **Supabase:** Built on standard PostgreSQL, making complex relationships easy to enforce. Its **Row-Level Security (RLS)** enforces backend access permissions natively, ensuring recruiters cannot access private student information. It also features native WebSockets for real-time notifications and direct messaging.
- **Google Gemini API:** Provides a single AI integration to power both the interactive onboarding chatbot and natural language resume processing (extracting skills from PDFs).

Real-World Scenario: A Student Applies for a Job

1. **Onboarding & CV Parsing:** A student registers using @my.richfield.ac.za. Upon initial login, **Gemini API** opens an interactive onboarding chat that assists with profile updates. The student uploads a resume PDF, which **Gemini** parses using natural language processing to populate their profile with extracted skills.
2. **Post Security & Verification:** An enterprise recruiter creates a business account. The account remains locked until a staff administrator approves the company via the **Supabase-backed Admin Web Panel**.
3. **Smart Job Matching & Live Applications:** The verified employer posts a software developer internship. A backend database query matches the job's required skills with student profile records.
4. **Real-Time Notification: Supabase Realtime (WebSockets)** pushes a live notification directly to matching students' devices. The student opens their **Flutter mobile app**, views the listing, and submits an application.